

	Jakub Chojna Senior Software Engineer 3d Graphic Artist Architect	Contact Info
	Full-stack TypeScript engineer with 6+ years of commercial experience building data-driven and visually rich web applications across React, Next.js, Remix and Node.js. My background in architecture and 3D graphics translates directly into work with UI design, Three.js/WebGL and interactive data visualization — areas I'm looking to deepen, particularly around digital twins and real-time 3D on the web. I've worked across startups and software houses, owning real-time features, design systems and performance-critical dashboards for Fortune 500 clients. I care about high-quality, maintainable code and polished interfaces, with a strong attention to visual detail and consistency.	✉ jakub.chojna@gmail.com 🐙 Github Repo 🏠 Portfolio 🌐 Linkedin
	Professional Experience	Technologies / Skills
Dec 2025 - Present	Senior Software Engineer Netfire LLC US Startup Remote	Programming Languages
	Improved application performance, scalability and Web Vitals by improving caching, memoization, eliminating layout reflows, lazy-loading routes and assets, and refactoring async data fetching — reducing unnecessary computation and client re-renders. Owned an ETL pipeline (AWS S3 → PostgreSQL → Shopify) handling multi-type product ingestion, schema validation, and automated lifecycle reconciliation across data stores. Maintained generated Zod schemas from OpenAPI specifications, enforcing type-safe API contracts and runtime validation across frontend and backend services. Implemented real-time AI chatbot functionality over WebSockets, including streaming responses, interactive workflows, and connection health monitoring. Drove a gradual data migration from a third-party commerce platform (Shopify) to PostgreSQL (Drizzle ORM), moving product queries and assets to database-backed resources without service interruption. Standardized error handling and improved observability across microservices with shared error types and unified response formats; expanded Sentry coverage to surface silent production failures.	TS TypeScript R R JS JavaScript Python Python Frameworks React Next.js Remix Electron Gatsby Shiny Backend & APIs Node Express.js GraphQL Rest API Deno FastAPI UI & Design Systems Tailwind CSS GSAP shadcn/ui styled-components Storybook HTML MUI Ant Design Sass
Jan 2025 - Dec 2025	Software Engineer Netfire LLC US Startup Remote	
	Developed full-stack features across a Remix frontend, Express.js microservices, and PostgreSQL database — including subscription flows, data-driven UI features, and API integrations end-to-end. Built and maintained reusable components within a shared styled-components design system (Storybook), ensuring consistent UI patterns across multiple applications. Implemented product listing interfaces with filtering, detail pages with dynamic layouts and data transformations, image galleries with resolution-optimized thumbnails, and loading skeletons. Ported authentication and content features between codebases, refactoring to fit different design systems and application patterns. Performed cross-browser functional and visual QA, then implemented responsive and accessibility fixes across the application. Prepared the application for search engine indexing — added sitemap generation, improved SEO, corrected URL structures, and removed non-public content from crawlable pages.	
Aug 2023 - Jan 2025	Senior Frontend Developer Appsilon Sp. z o. o. Software House Remote	
	Designed scalable frontend architecture for real-time data dashboards serving Fortune 500 clients in Pharma, Biodiversity, Energy, and Sports — handling continuous data streams, complex state, and heavy visualization workloads (polling, SSE, WebSockets) Led small-scale projects end-to-end, owning technical direction, code reviews, and coordination within distributed international teams of frontend, backend, DevOps engineers, and data scientists. Developed APIs using Python/FastAPI and R/Plumber, integrated with React/Next.js frontends. Deployed and maintained containerized applications using Docker on AWS EC2 instances. Built advanced data visualizations and 3D interactive experiences using D3.js, Three.js, and WebGL, optimizing rendering performance for data-intensive views. Built an interactive supply chain delivery flow visualization using Next.js, React Flow and Neo4j database, mapping complex graph relationships across logistics networks within an analytical dashboard. Established testing practices across projects — TDD workflows, unit testing (Jest/Vitest), and E2E coverage (Cypress, Playwright), including visual regression pipelines. Built and maintained cross-product design systems with Storybook, Tailwind CSS, and component libraries (shadcn/ui, Material UI, Ant Design), standardizing patterns across teams.	3D & Visualization Three.js d3.js WebGL Echarts Recharts Databases & Data Zod Drizzle ORM Prisma ORM PostgreSQL Neo4j

Jun 2021 - Aug 2023	Frontend Developer Appsilon Sp. z o. o. Software House Remote
	Shared knowledge and best practices on UI/UX design and frontend development across company projects, helping teams improve consistency and overall quality of user interfaces. Developed interactive dashboard applications displaying continuously updated analytical data, integrating REST APIs and charting libraries within React/TypeScript and R Shiny codebases. Contributed to the development of internal frameworks and libraries bridging React, Next.js, Tailwind with the R/Shiny ecosystem, enabling reusable components and modern frontend patterns within Shiny applications. Built standalone desktop applications with Electron and static sites with Gatsby. Implemented animated interfaces and UI transitions using GSAP, Three.js, improving usability and engagement on client-facing products. Created wireframes, prototypes, and UI concepts in Figma, collaborating with designers and stakeholders from early product stages through delivery. Wrote unit and integration tests as part of daily development, contributing to team-wide quality standards.
Jun 2020 - Jun 2021	Junior Frontend Developer Appsilon Sp. z o. o. Software House Remote
	Built responsive, cross-browser-compatible dashboard UIs, raising the baseline for visual quality and consistency on client projects. Contributed to the company website (https://www.appsilon.com/), marketing landing pages, open-source initiatives and wrote technical blog posts. Created UI/UX designs for dashboards in Figma myself or collaborated with the design team during early project phases. Built and maintained a shared component library, improving development velocity and UI consistency across projects.
2008 - 2020	3d Graphic Artist Remote / Warsaw, Poland
	Created photorealistic architectural visualizations of interiors and exteriors for clients including Echo Development, Terroir, Konior Studio, Studio Thörnblom, Real2B Development, Mallson, MOFO Architekci. Created walkthroughs using Unreal Engine leveraging real-time rendering. Conducted several 3ds Max and V-Ray trainings for Euro Info Group Sp. z o.o., covering 3D modeling, texturing, lighting, rendering, architectural visualization, post-production, animation, and particle systems. Architectural Visualizations Portfolio
2011 - 2018	Architect Remote / Warsaw, Poland / Ljubljana, Slovenia
	Developed concept, schematic and construction documentation, produced 3D models, 2D graphics, and photorealistic architectural visualizations in Svet Vmes d.o.o., Mofo Architekci and Activum. Developed furniture prototypes and contributed to the refurbishment of a single-family house in Lublin in collaboration with architect Piotr Gniewek. Architectural Portfolio
	Education
	Faculty of Architecture Warsaw University of Technology Master Degree
	Faculty of Architecture Warsaw University of Technology Bachelor Degree
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Libraries & Tools Git TanStack Vite.js Redux Zustand i18next Cloud & Infrastructure AWS Docker Shopify Testing Jest Vitest Cypress Playwright UI & Graphic Design Figma Photoshop Illustrator After Effects 3d Graphics 3ds Max V-Ray Rhinoceros Blender Unreal Engine Revit Autocad
Interests 3D Printing Wood-working Computational Design Race Cycling
Languages Polish native English proficient

Spanish

German